

INDIAN STATISTICAL INSTITUTE
(BSDS)_(2026-2027), MID-SEMESTER EXAMINATION
(Signal, Image & Text Processing)

PART-1 and PART-2 (30 marks each). Final mid-sem mark will be scaled to 30 marks

PART-1

Duration: 60 minutes

Maximum Marks:30

Note: Answer any TWO

1.
 - a. What is a signal, and why does it demand processing? [3]
 - b. Which type of signal is preferred in present-day applications? Analog, digital, or mixed? Justify. [4]
 - c. Provide a mathematical representation of Linear, Quadratic, exponential, and sinusoidal signals. [4]
 - d. Name and classify the following signals according to whether they are (1) 1D or [2+2]
multi-dimensional; (2) single or multichannel, (3) continuous time or discrete time, and
(4) analog or digital (in amplitude)
 - i. A child's weight and height are measured every month.
 - ii. A colour television picture
2.
 - a. Perform the linearity check of the systems [6]
 - i. An upsampler
 - ii. $y(n) = Mx(n) + C$
 - b. Perform a time-variance check of the systems [6]
 - i. A downsampler
 - ii. $y(n) = x(-n + 2) + x(n - 3)$
 - c.
 - i. What is the passivity of a digital system? [3]
 - ii. Under what conditions is the following system
 $y(n) = \alpha x(n - N)$, (α is an arbitrary constant) passive and active?
3.
 - a. Determine the power and energy of the unit step sequence, and check if it is a power or energy signal. [4]
 - b.
$$x(n) = \begin{cases} 1 + (n/3), & -3 \leq n \leq -1 \\ 1, & 0 \leq n \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$
 [5]

Find $x(-4 - n)$, and $x\left(-\frac{n}{2}\right)u(n + 3)$
[Any format of the representation is fine]
 - c. Signal $x(n) = \{2, 3, 1, 4, -5\}$, for $n = \{-2, -1, 0, 1, 2\}$. [6]
Find EVEN and ODD signals of $x(n)$ and combine them to get the original signal.

===== End of Question paper for PART-1 =====

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PART-2

Duration: 60 minutes

Maximum Marks:30

Note: Answer any TWO

- 4.
- Find the inverse system of the system $y(n) = \sum_{k=0}^{\infty} x(n-k)$ [5]
 - Find the values of the sequence $x(n) = u(n) + u(1-n)$ and represent the result in terms of $\delta(n)$ in the most possible simplified form. [2]
 - Find the values of the sequence $x(n) = u(n) + u(-n)$ and represent the result in terms of $\delta(n)$ in the most possible simplified form. [2]
 - Find the impulse and unit step responses of the following system [6]

$$y(n) = x(n) + \frac{1}{2} [x(n-1) + x(n+1)]$$

- 5.
- Justify: All continuous-time signals are Analog, but not vice versa [3]
 - For the input signal: [3]

$$x(n) = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Determine the response of the following system.

$$y(n) = \sum_{k=-\infty}^n x(k) = x(n) + x(n-1) + x(n-2) + \dots \text{ (accumulator)}$$

- What do you say about the stability of an accumulator? Is it stable or unstable, and under what conditions? [3]
 - What is the aliasing effect in digital signal processing? Given a combined digital signal of THREE individual signals, $\cos 6\pi n$, $\cos 14\pi n$, and $\cos 26\pi n$, check and provide your views on the advantage/disadvantage of the sampling frequency at 10 Hz and 50 Hz for the combined signal. [6]
- 6.
- Prove the commutative property of the convolution operation. [3]
 - Analytically and graphically find the convolution of an input sequence $x(n) = \{1, 2, 3, 0, 4\}$, for $n = \{-1, 0, 1, 2, 3\}$ with the impulse response of the system $h(n) = \{2, 1, 4, 6\}$, for $n = \{-2, -1, 0, 1\}$. [6]
 - Represent a combined system that combines two individual systems in cascade and parallel. Also, comment on the stability of the cascade and parallel realizations. [3]
 - Represent the different realizations of the system called the accumulator. [3]